Vansh Chugh

vchugh@knox.edu | LinkedIn | GitHub | ChughVansh.github.io

Research Interests

Audio effects representation learning, real-time human–AI music interaction, democratization of MIR technologies

Education

B.A. in Data Science; Minors in Mathematics & Pop Music Sep. 2023 – June 2027

Knox College | GPA: 4.0/4.0 *Galesburg, IL*

- **Senior Honors Thesis (in progress):** Real-Time Accompaniment Generation for Human–AI Co-Creation. Advisors: Benjamin Penwell, Ole J. Forsberg
- **Junior Honors Thesis** (sole junior admitted to Honors Program): Context-Aware Representation Learning for Mix Compatibility Estimation. Advisor: Ole J. Forsberg. External examiner: Michael Clemens (Universal Audio)

Publications

V. Chugh and Y.-T. Yeh, "Mix-JEPA: A Joint Embedding Predictive Architecture for Music Mixing Representation," under review at *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2027.

V. Chugh, A. Phan, and K. Fatima, "Oh Sheet!: A Browser-Based Optical Music Recognition Tool for Classical and Handwritten Jazz Lead Sheets," in *Extended Abstracts for the Late-Breaking Demo Session of the 27th International Society for Music Information Retrieval Conference (ISMIR)*, 2026.

V. Chugh, "Prompt-Based Music Discovery: A Prototype Using Source Separation and LLMs," in *Extended Abstracts for the Late-Breaking Demo Session of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025; also presented at the *NeurIPS 2025 Workshop on Artificial Intelligence for Music: Where Creativity Meets Computation (AI4Music).*

Research Experience

Research Intern June 2026 – Sep. 2026

Audio Information Research Lab, University of Rochester | Advisor: Zhiyao Duan | HARP | Link

- Deployed 25 of HARP's 75 models in 3 months, pairing LLM-agent judgment with deterministic verification gates (dead-code pruning, preflight checks, live inference probing), cutting deployment time from hours to minutes
- Contributed to a systematic validation framework testing models across varied input lengths, sample rates, channel counts, and file types (95% of Spaces passing), and reviewed two 4,000+ line PRs, including the framework itself
- Built a JUCE C++ text-output component, extending output support beyond audio and MIDI files

Project Lead Mar. 2026 – June 2026

Oh Sheet!: Optical Music Recognition Web App | Link

- Built a browser-based sheet music transcription app that converts photos and scans into editable, playable MusicXML files, reaching 40+ registered users and 300+ model calls in 2 months at $0 hosting cost
- Engineered a serverless GPU inference pipeline on Modal (A10G) with dynamic routing across a classical LLM, CNN, and jazz-specific YOLO+sequence model by score type, keeping per-call cost to ≈$0.01
- Identified and integrated a 2025 ISMIR jazz transcription model after user interviews revealed systematic failure on handwritten lead sheets, enabling a capability absent from competing browser-based tools

Knox Summer Scholar June 2025 – Aug. 2025

Advisor: Ole J. Forsberg | Whatcha Lookin' For: Prompt-Based Music Recommendation | Link

- Built a full-stack natural-language music recommendation system (FastAPI, Groq-hosted LLaMA 3.3 70B, SQLite) that translates prompts like "metal songs with violin" into database queries via schema-aware prompt engineering for safe LLM-driven SQL generation
- Developed a scalable ETL pipeline, including BeautifulSoup web scraping, Demucs-based source separation, API integration, and librosa heuristics to extract instrument-presence features for 650+ tracks

Professional Experience

Data Scientist / Backend Intern Aug. 2025 – Sep. 2025

EdPi Innovate | Link

- Built the company's single-source, configurable meeting transcription and summarization system using Whisper, FastAPI, and SpeechBrain, reducing documentation time by 80% and improving organizational knowledge capture
- Engineered a speaker identification pipeline using diarization, voice embeddings, and cosine-similarity matching against employee voice profiles, enabling 95%+ accurate, fully automated speaker labeling for 50+ meetings/month
- Developed an LLM-driven summarization and action-item extraction module that maps tasks to emails, improving follow-through by 60% and reducing admin workload by 3–4 hours/week through automated follow-up processing

Data Analyst Intern Nov. 2024 – Jan. 2025

Knox College, Office of Advancement

- Analyzed 50+ years and $50M+ in alumni donation data using non-parametric statistical tests and inflation-adjusted donor metrics to evaluate cohort and involvement-based differences, informing the 2025 segmentation strategy

Data Management Intern Nov. 2023 – Jan. 2024

Joe Moore Strategies, LLC

- Designed a Google Sheets tracking system for 30,000+ clients by position, engagement level, and relationship to the company, migrating 1,500+ government-official contacts to Monday.com, reducing data retrieval time by 60%

Projects

Statistical Hypothesis Testing | *Monte Carlo Study of Normality Tests* | Link Mar. 2025 – June 2025

- Showed via parameter sweeps that Shapiro–Wilk holds nominal significance levels across sample sizes while Anscombe–Glynn under- and over-rejects, and that Shapiro–Wilk better detects asymmetry under heavy tails (t, χ^2 alternatives)

NBA Game Outcome Predictor | Link Jan. 2025 – Mar. 2025

- Engineered features from 20+ game statistics and tuned Logistic Regression, Random Forest, XGBoost, and neural network models with GridSearchCV while preventing data leakage, reaching up to 64% accuracy; documented in a 13-page paper

Talks & Presentations

"Automatic Mixing, FX Representations, and the WIMIR Mentee Experience," WIMIR June 2026

"Google, Recommend Some Death Metal with Acoustic Guitar: Prompt-Based Music Discovery with Source Separation and LLMs"

HORIZONS Research Symposium, Knox College May 2026
Stanford Deep Learning for MIR Summer Workshop (instructors: Iran Roman, Kitty Shi) Aug. 2025
Knox Summer Scholars, Knox College July 2025

Honors & Awards

Election to Phi Beta Kappa (sole junior inductee) 2026
College Marshal 2026
Lucile Sudbury Prize, Best Scholarly Paper in Music Theory/Analysis ($1,000) 2026
Richter Memorial Fund ($3,000; $1,200; $900; $750) 2023–2026
ISMIR Registration Waivers ($250; $300); NeurIPS Workshop Student Grant ($200) 2025–2026
ASA DataFest Awards (Best Insight; Best Use of External Data; Honorable Mention) 2024–2026
Trustees Distinguished Scholarship ($31,000/year, 4×) 2023–2026
Knox Summer Scholars Program ($5,000; largest research grant at Knox) 2025
Clark Mills Carr Prize in Mathematics (1st, college-wide competition) 2025
Sigma Xi Best Student Seminar ($100) 2025
Dean's List (7×) 2023–Present

Teaching, Mentoring & Service

Tutoring Captain, Knox College Jan. 2024 – Present
Most-booked math tutor; 800+ hours of 1:1 tutoring in Calculus I–III, Linear Algebra, Discrete Mathematics, Introduction to Statistics, Mathematical Statistics I–II, Data Mining, and Linear Models

Research Ambassador, Knox College Sep. 2025 – Present
Advised 40+ student researchers on grant applications through 1:1 proposal reviews; supported classroom visits and events promoting undergraduate research

Teaching Assistant, Linear Algebra, Knox College Jan. 2026 – Mar. 2026
Guided 20+ students, graded 200+ assignments, and held 6+ hours of office hours weekly

Musical Background

Independent Artist | *1shh* | Link Sep. 2019 – June 2022

- Produced, mixed, and mastered original tracks released on major streaming platforms, leading end-to-end production from ideation and recording through cover art, videography, and promotion to reach 13,000+ YouTube views in 10+ countries
- Performed live with Arijit Anand (200K+ monthly listeners) at a concert with 500+ attendees

Tabla, 6th Year Examination, Prayag Sangeet Samiti Mar. 2018

Technical Skills

Languages: Python, C++, R, SQL
ML & Audio: PyTorch, librosa, Demucs, Whisper, SpeechBrain, scikit-learn, TensorFlow, XGBoost
Audio Software & Production: JUCE, Max/MSP, Logic Pro, FL Studio
Tools & Infrastructure: Modal, Hugging Face Spaces, FastAPI, Git, LaTeX, SQLite